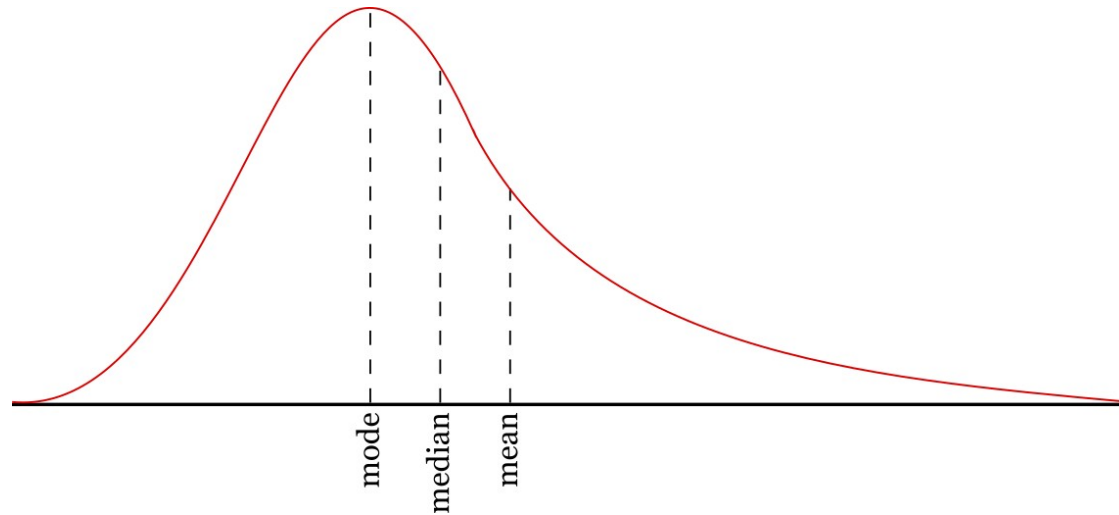


Lecture 9

Descriptive Statistics II



Announcements

- ❖ Lab 3 this Wednesday
 - Using aggregation to analyze a dataset
- ❖ Course project details coming soon
- ❖ Grades for labs will be up this week

Last Time

- ❖ What is a statistic?
- ❖ Histograms to summarize the "distribution" of a variable
- ❖ Central tendency
 - Mean & median
- ❖ Position
 - Quantiles
- ❖ Spread
 - Variance and standard deviation

Today

- ❖ A few more statistics
 - Mode – most common value
 - Covariance & correlation coefficients
- ❖ Analyze debugging data

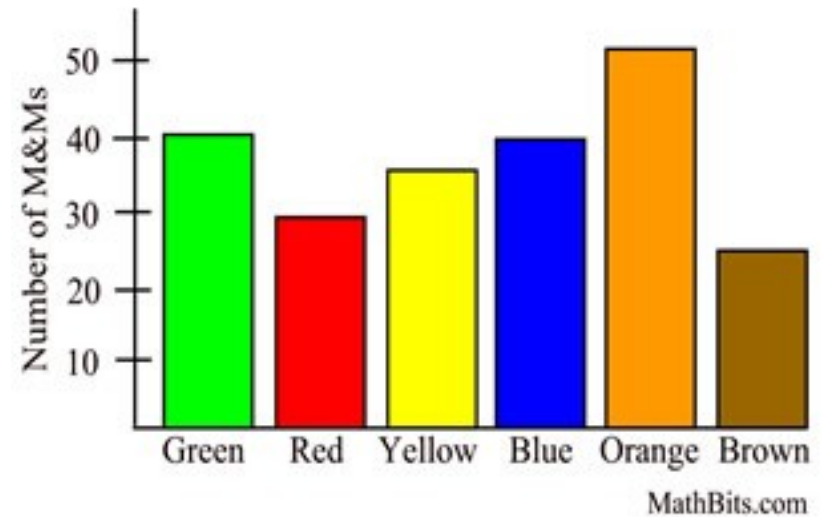
What is a statistic?

- ❖ A summary of the data
- ❖ Often captures one aspect or feature of a dataset

Counts & Histograms

How many of each value are in our dataset?

- ❖ Gives us a full summary of the variable in our dataset



Number of Colors in Bag of M&M Candies

Centrality

- ❖ For variables that can be ordered, it's helpful to know a typical value

$$X = [X_1, X_2, X_3, \dots, X_n]$$

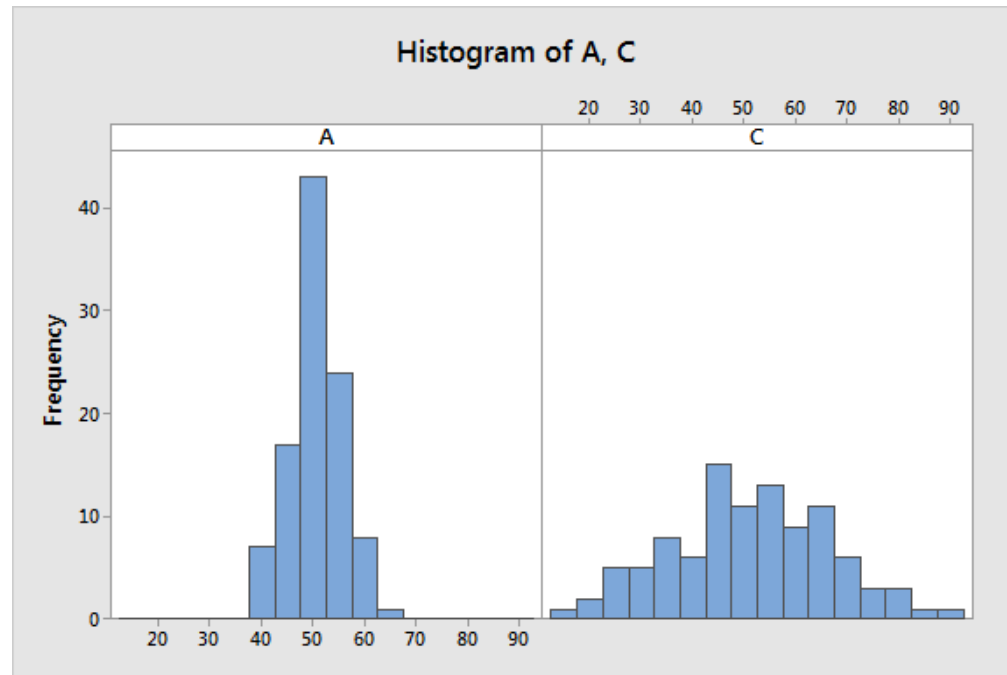
- ❖ Mean (or average)
 - Sum all the values and divide by n ; denoted by $\bar{x}$
- ❖ Median
 - Sort the values and find the middle value (or average the two middle values)

Position

- ❖ The median is the middle value
 - 50% of the dataset has a value less than or equal to the median
- ❖ A quantile generalizes the median to any %
 - Sometimes called a percentile as well
- ❖ Multiple quantiles can summarize a variable distribution
 - Common quantiles: 0%, 25%, 50%, 75%, 100%

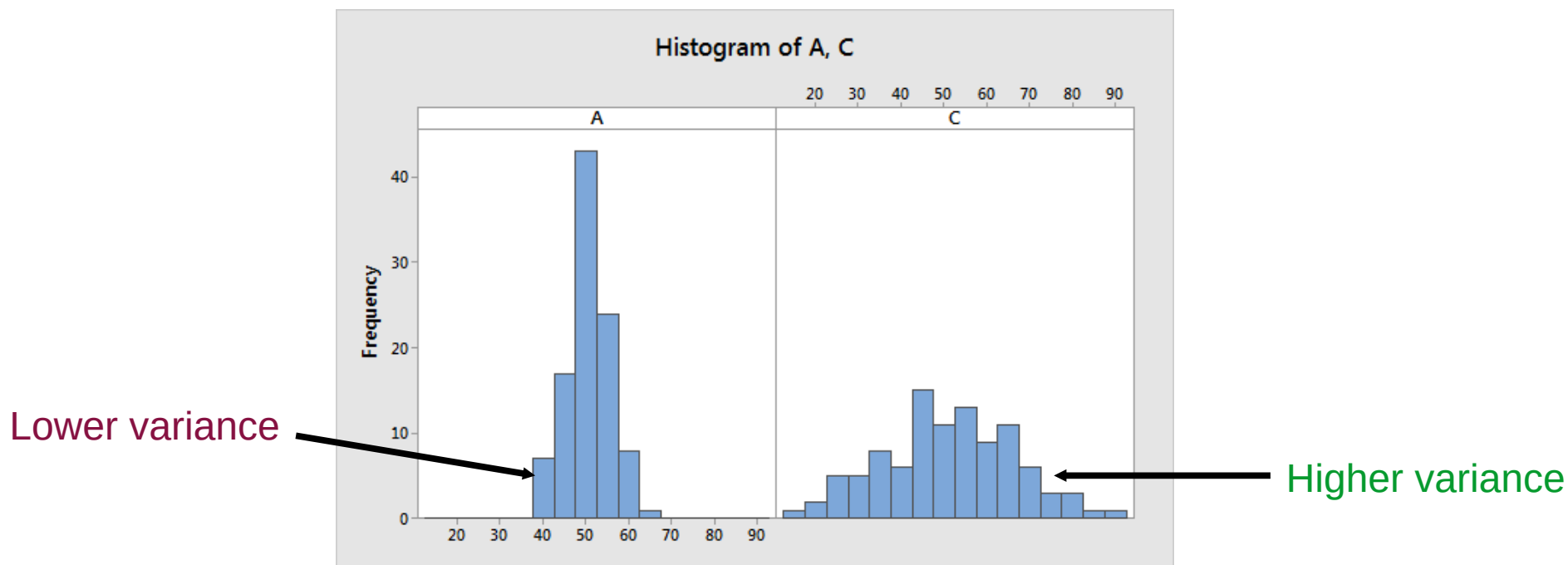
Spread

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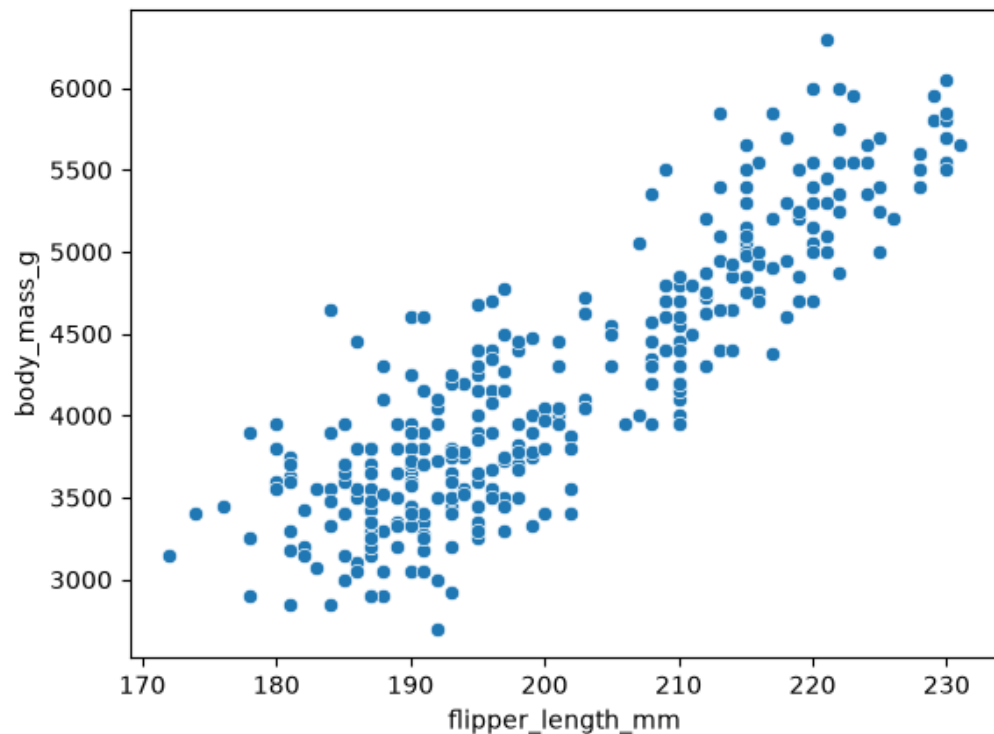
Spread

- ❖ Variance is a measure of how far the values in the dataset are from the mean
- ❖ Standard Deviation is the square root of variance

x	$\bar{x}$	$Var(x)$	$SD(x)$
Variable	Mean	Variance	Standard Deviation

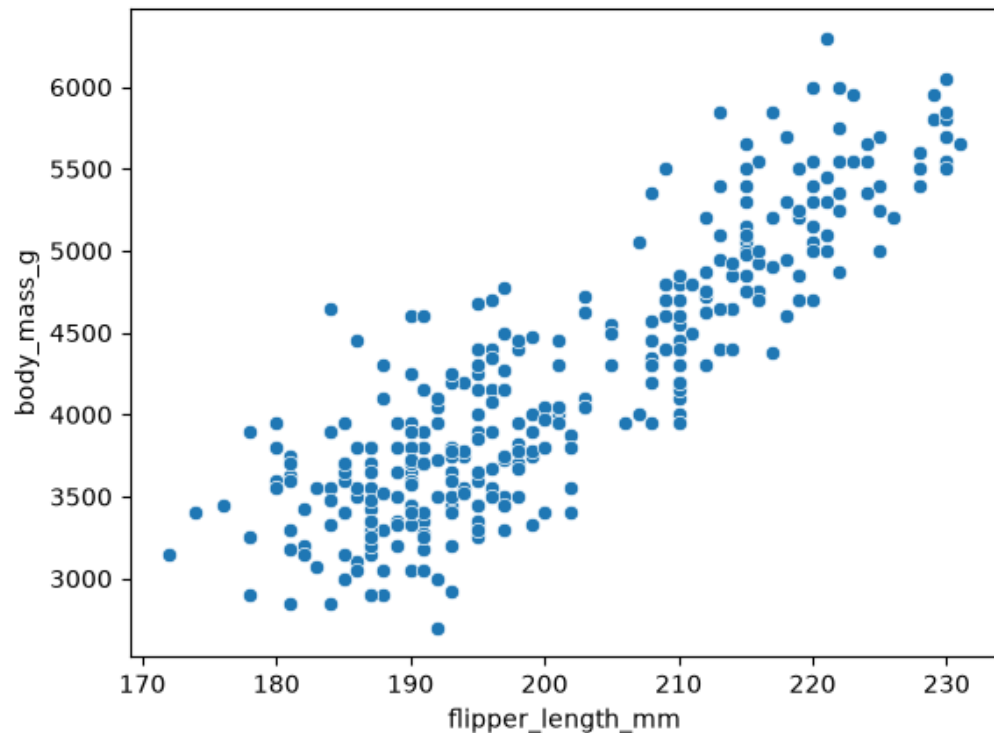
Relationships

- ❖ How would you describe this relationship?



Relationships

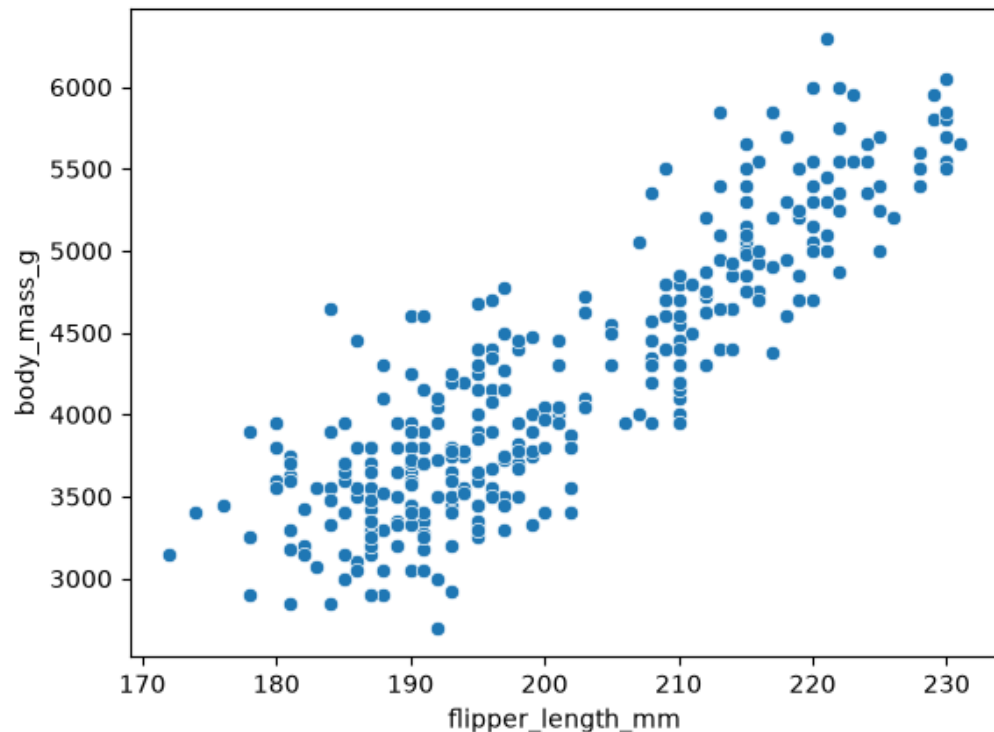
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- ❖ **Covariance** is the statistic that captures this idea



Relationships

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$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

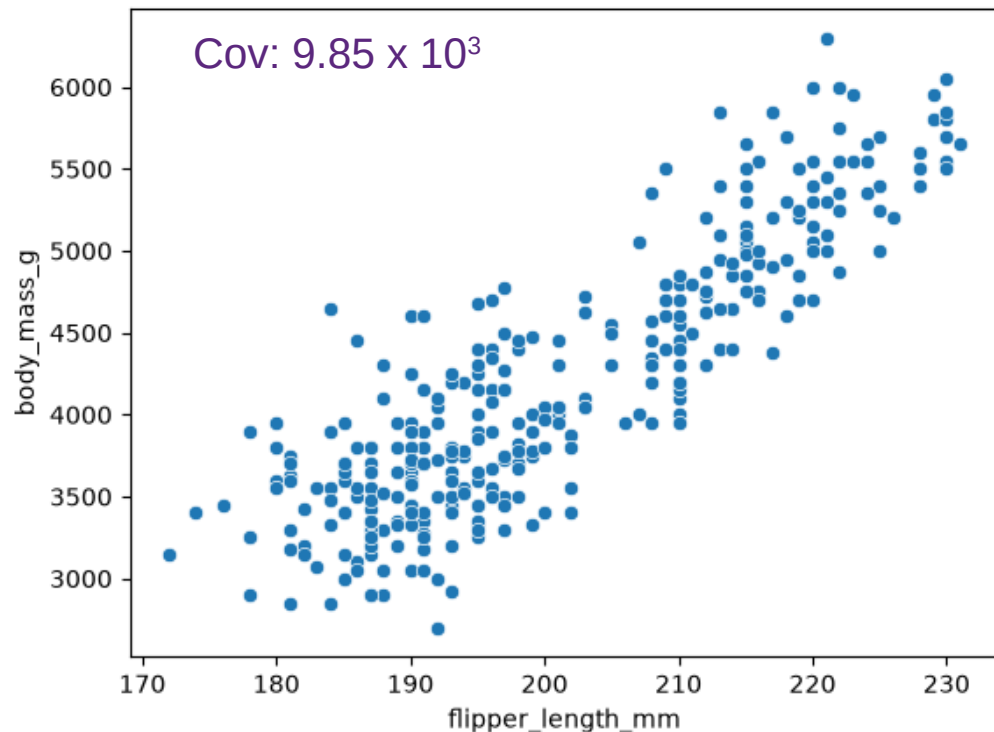


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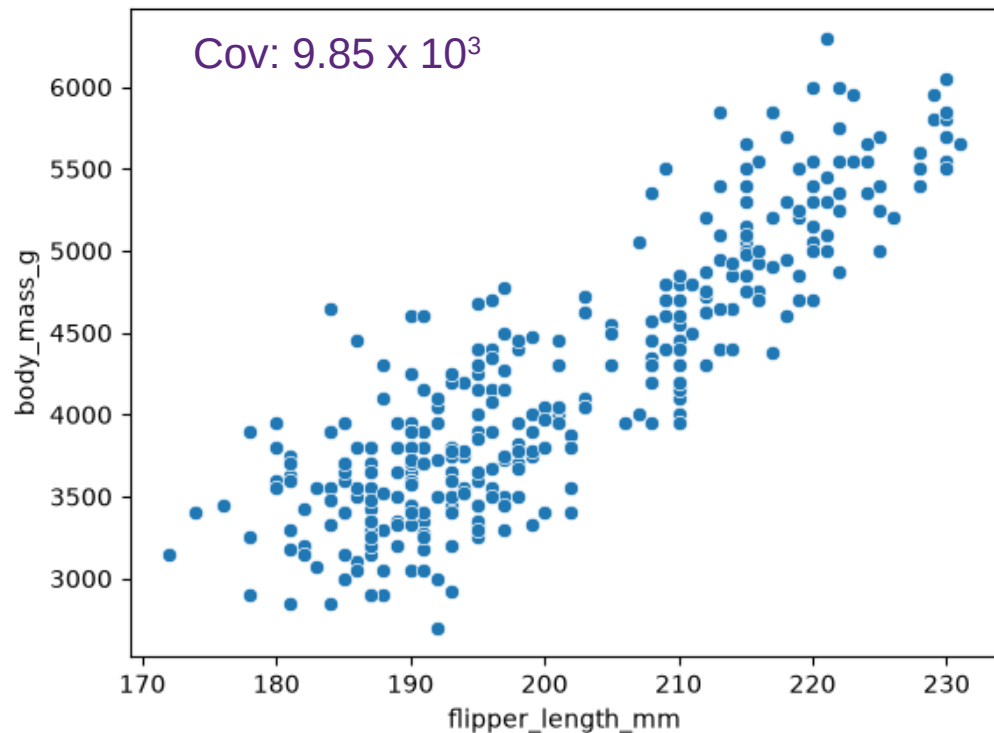
$$\text{Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- ❖ > 0 -> positive relationship
- ❖ < 0 -> negative relationship



Relationships

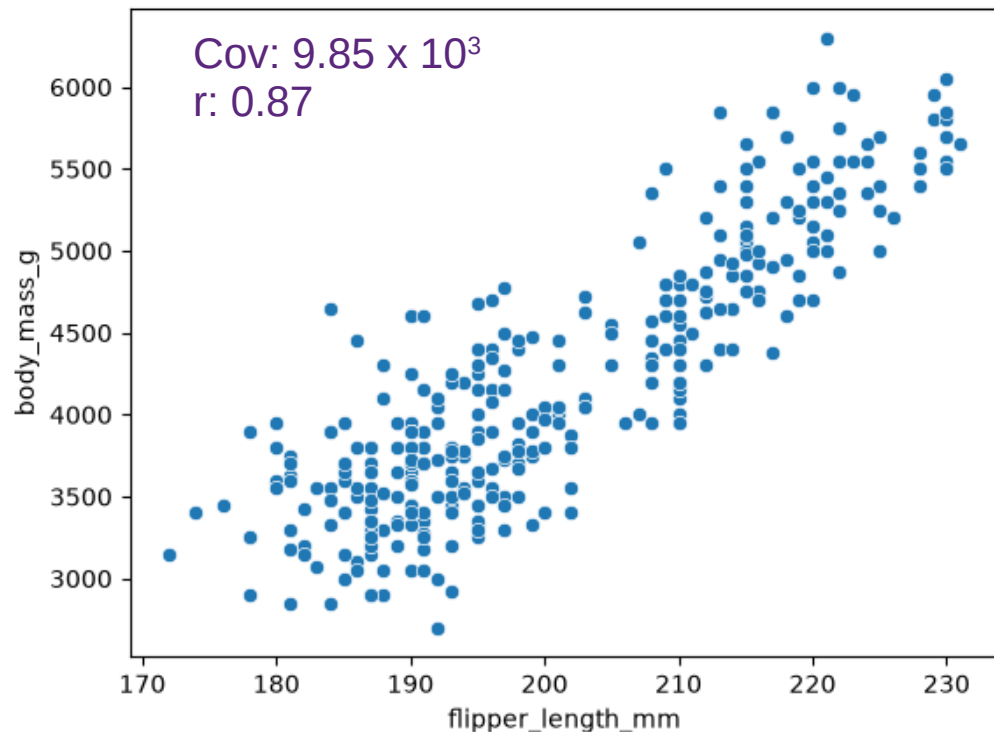
- ❖ Covariance as a number is difficult to interpret



Relationships

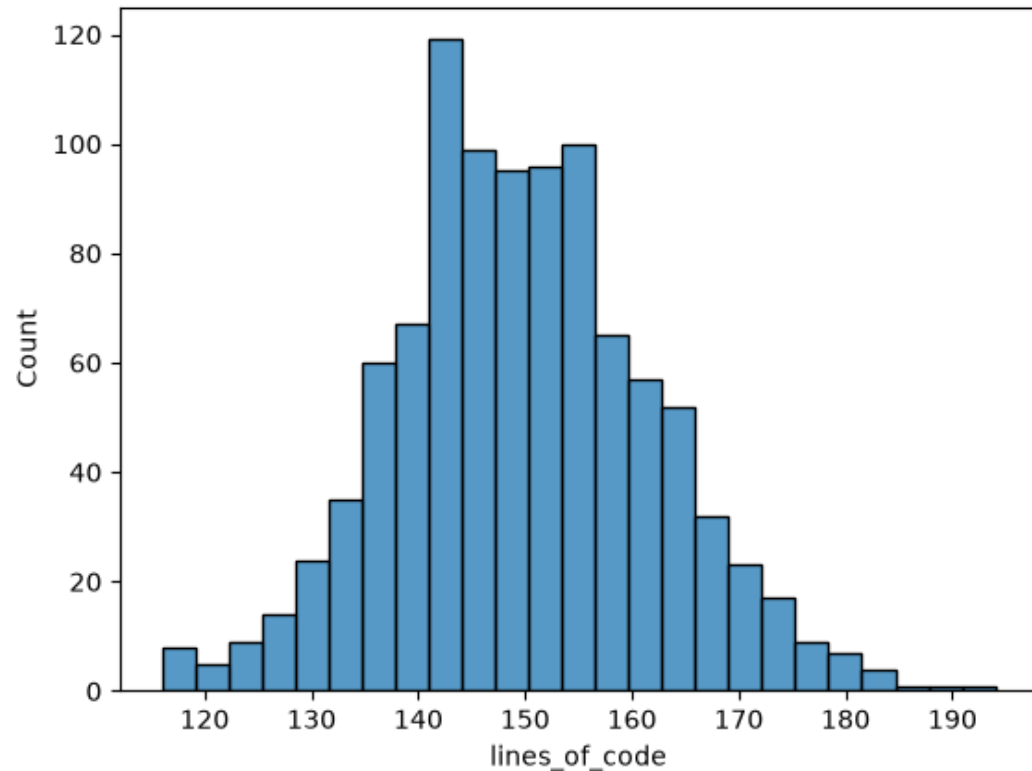
- ❖ Covariance as a number is difficult to interpret
- ❖ We can get obtain a unitless quantity between -1 and 1 by dividing covariance by SDs
- ❖ We call this a **correlation coefficient**

$$r(x, y) = \frac{\text{Cov}(x, y)}{\text{SD}(x) \text{SD}(y)}$$



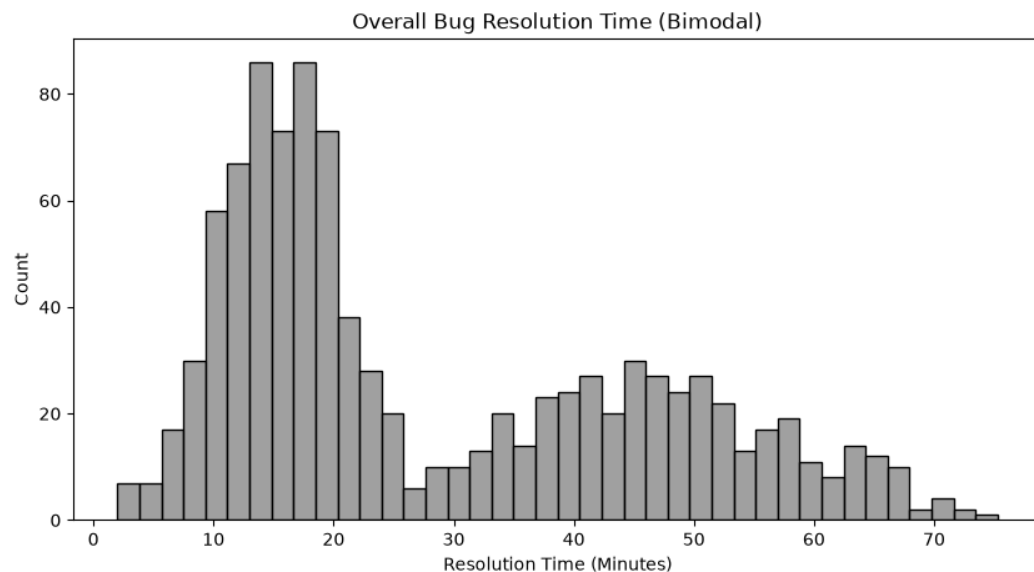
Mode

- ❖ Most common value in attribute
- ❖ Useful for describing distributions



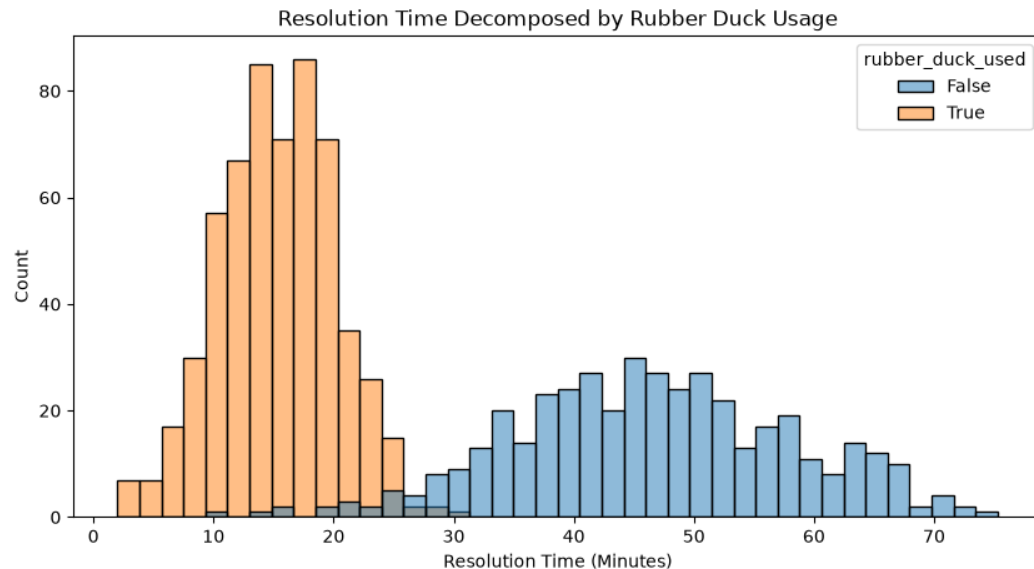
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- ❖ Call this "bimodal"



Mode

- ❖ Most common value in attribute
- ❖ Useful for describing distributions
- ❖ Call this "bimodal"
- ❖ Often bimodality indicates an interesting trend in the dataset



Next Time

- ❖ Read DS Ch. 3
- ❖ Aggregating Data
 - Using groupby in Python